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Pulse noise modulation to encode data

Abstract

A method, apparatus, and system for communicating data. Data is identified for transmission. Pulses of noise signals encoding the data are transmitted.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1309031	12/1918	Hettinger	N/A	N/A
3404403	12/1967	Vallese et al.	N/A	N/A
3719829	12/1972	Vaill	N/A	N/A
6087992	12/1999	Anderson	N/A	N/A
6087993	12/1999	Anderson et al.	N/A	N/A
6377436	12/2001	Margolin	N/A	N/A
6650297	12/2002	Anderson et al.	N/A	N/A
6657594	12/2002	Anderson	N/A	N/A
6674970	12/2003	Anderson	N/A	N/A
7456791	12/2007	Pellet	N/A	N/A
7613074	12/2008	Blackmon et al.	N/A	N/A
7965241	12/2010	Marquis	N/A	N/A
8922436	12/2013	Stone et al.	N/A	N/A
10069564	12/2017	Hening et al.	N/A	N/A
10211522	12/2018	Anderson	N/A	N/A
10630383	12/2019	Zhang	N/A	H04B 10/0795
10763963	12/2019	Jia et al.	N/A	N/A
11024950	12/2020	Hening et al.	N/A	N/A
2007/0215946	12/2006	Eastlund	N/A	N/A
2014/0288539	12/2013	Bischoff	606/4	A61F 9/00825
2018/0263103	12/2017	Exton	N/A	B64C 23/005
2019/0094185	12/2018	Athanassiadis	N/A	N/A
2019/0379459	12/2018	Kim	N/A	H04B 10/564
2022/0390553	12/2021	Englesbe	N/A	G01S 13/0209
2024/0244738	12/2023	Walczak	N/A	H05H 15/00
2024/0267126	12/2023	Howe	N/A	H04B 10/504

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111756470	12/2019	CN	H04W 12/02
124833	12/1918	GB	N/A
WO-2010111089	12/2009	WO	B23K 26/40
2017/210871	12/2016	WO	N/A

OTHER PUBLICATIONS

Antipov A A et al: “Microwave Generation in an Optical Breakdown Plasma Created By Modulated Laser Radiation”, Soviet Journal of Quantum Electronics, American Institute of Physics, New York, NY, US, vol. 6, Jun. 1, 1990 (Jun. 1, 1990), pp. 664-666, XP000174924, ISSN: 0049-1748, DOI: 10.1070/QE1990V020N06ABEH006676. cited by applicant

European Patent Office Extended Search Report, dated Apr. 25, 2024, regarding Application No. EP23205184.7, 10 pages. cited by applicant

Griffiths et al., “Scalable visible light communications with a micro-LED array projector and high-speed smartphone camera,” Optics Express, vol. 27, No. 11, May 27, 2019, pp. 15585-15594, <https://opg.optica.org/oe/fulltext.cfm?uri=oe-27-11-15585&id=412698>. cited by applicant

Li et al., “Optical Computing using hybrid encoded shadow casting,” Rapid Communications, Applied Optics, vol. 25, No. 16, Aug. 15, 1986, 4 pages. cited by applicant

Zong Si-Guang et al: “Study of opto-acoustic communication between air and underwater carrier”, Proceedings of SPIE; [Proceedings of SPIE ISSN 0277-786X vol. 10524], SPIE, US, vol. 10697, Feb. 20, 2018 (Feb. 20, 2018), pp. 106975S-106975S, XP060102900, DOI: 10.1117/12.2315446, ISBN: 978-1-5106-1533-5. cited by applicant

Kumar et al., “Detection and analysis of low-frequency electromagnetic emissions from ns laser induced breakdown of air,” Proceedings of SPIE—The International Society for Optical Engineering, May 2012, vol. 8434, 7 pages. cited by applicant

Mechain et al., “A virtual antenna produced in air by intense femtosecond laser pulses,” European Symposium on Optics and Photonics for Defence and Security, Nov. 3, 2005, Bruges, Belgium, 7 pages. <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/5989/59890S/A-virtual-antenna-produced-in-air-by-intense-femtosecond-laser/10.1117/12.631202.short>. cited by applicant

Brelet et al., “Radiofrequency plasma antenna generated by femtosecond laser filaments in air,” Applied Physics Letters, vol. 101, Issue 26, Dec. 27, 2012, 4 pages. <https://aip.scitation.org/doi/abs/10.1063/1.4773492>. cited by applicant

Theberge et al., “Broadband and long lifetime plasma-antenna in air initiated by laser-guided discharge,” Applied Physics Letters, vol. 111, Issue 7, Aug. 14, 2017, 5 pages. <https://aip.scitation.org/doi/10.1063/1.4985045>. cited by applicant

Alshershby et al., Reconfigurable Plasma Antenna Produced in Air by Laser-induced Filaments: Passive Radar Application, 2012 International Conference on Optoelectronics and Microelectronics (ICOM), Aug. 23-25, 2012, Changchun, China, 8 pages. cited by applicant

Choe et al., “Radiation characteristics of femtosecond laser-induced plasma channel Vee antenna,” Physics of Plasmas, vol. 22, Issue 6, 10.1063/1.4922083, Published Online Jun. 4, 2015, 10 pages. cited by applicant

Stallings, “Spread Spectrum,” Data and Computer Communications 7th Edition, Chapter 9, Feb. 2007, 28 pages. <http://williamstallings.com/DCC/DCC7e.html>. cited by applicant

Dai et al., “Demonstration of 17 Meter Standoff THz Wave Generation,” Nonlinear Optics: Materials, Fundamentals and Applications 2009, Honolulu, Hawaii, Jul. 12-17, 2009, 4 pages. cited by applicant

Wang et al., “Effect of laser energy and polarization on RF emission characteristics of laser induced

air plasma,” Eleventh International Conference on Information Optics and Photonics (CIOP 2019), 2019, Xi'an, China, Proceedings of SPIE vol. 11209, 8 pages.
<https://www.spiedigitallibrary.org/conference-proceedings-of-spie>. cited by applicant
Kumar et al., “Effect of laser intensity on radio frequency emissions from laser induced breakdown of atmospheric air,” Journal of Applied Physics, vol. 119, Issue 21, 214904, Jun. 7, 2016, 9 pages.
<https://doi.org/10.1063/1.4953211>. cited by applicant
Pearlman et al., “Emission of rf radiation from laser-produced plasmas,” Journal of Applied Physics, vol. 49, Issue 1, 457 (1978), Published Online Aug. 12, 2008, 4 pages.
<https://doi.org/10.1063/1.324360>. cited by applicant
Harilal et al., “Lifecycle of laser-produced air sparks,” Physics of Plasmas, vol. 22, Issue 6, 10.1063/1.4922076, Published Online Jun. 3, 2015, 35 pages.
<https://aip.scitation.org/doi/10.1063/1.4922076>. cited by applicant
Kumar et al., “Radio frequency emissions from laser induced material breakdown,” 10th Asia Plasma and Fusion Association Conference, International Conference Series, Journal of Physics, Conference Series 823, (2017), 012008, 7 pages. cited by applicant
Kumar et al., “RF emissions from laser breakdown of target materials,” 2013 International Conference on Microwave and Photonics (ICMAP), Dec. 13-15, 2013, 4 pages. doi: 10.1109/ICMAP.2013.6733512. cited by applicant
Kumar et al., “Spectral selective radio frequency emissions from laser induced breakdown of target materials,” Applied Physics Letters, vol. 105, Issue 6, 10.1063/1.4893279, Published Online Aug. 13, 2014, 6 pages. <https://doi.org/10.1063/1.4893279>. cited by applicant
Kumar et al., “A reconfigurable plasma antenna,” Journal of Applied Physics, vol. 107, Issue 5, 10.1063/1.3318495, Journal of Applied Physics > vol. 107, Issue 5, 10.1063/1.3318495, 10 pages. <https://doi.org/10.1063/1.3318495>, Mar. 2010. cited by applicant
Howe et al., “Radio Frequency Communications Using Laser Optical Breakdowns,” filed Dec. 16, 2022, U.S. Appl. No. 18/067,516, 100 pages. cited by applicant
European Patent Office, Extended Search Report, dated Nov. 11, 2024, regarding Application No. EP24180478.0, 12 pages. cited by applicant
European Patent Office, Extended Search Report, dated Oct. 25, 2024, regarding Application No. EP24180480.6, 10 pages. cited by applicant
Office Action, dated Apr. 23, 2025, regarding U.S. Appl. No. 18/334,739, 87 pages. cited by applicant
Office Action, dated May 14, 2025, regarding U.S. Appl. No. 18/067,516, 44 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is related to the following U.S. Patent Application entitled “Radio Frequency Communications Using Laser Optical Breakdowns”, Ser. No. 18/067,516, filed Dec. 16, 2022, assigned to the same assignee, and incorporated herein by reference in its entirety.

BACKGROUND INFORMATION

1. Field

(2) The present disclosure relates generally to communications and in particular, to methods,

apparatuses, systems, and computer program products for communicating data using radio frequency, optical, and/or other signals within the electromagnetic spectrum without physical antenna structures.

2. Background

(3) Wireless communications using radio frequency signals, optical, and/or other signals within the electromagnetic spectrum are common and widespread. Radio frequency signals are commonly used in computer networks, for example, in the form of Wi-Fi signals that provide communications links between various computing devices.

(4) Radio frequency signals are also used for communications between various clients such as ships, aircraft, land vehicles, buildings, and other locations. These communications can include data such as position information, voice messages, voice communications, and other types of data. For example, other types of data can include digital and analog signaling.

(5) Communications using radio frequency transmissions are facilitated using physical antennas. The transmission or reception of radio frequency signals occurs between antennas. The use of physical antennas can be less convenient or reliable than desired.

(6) In addition, radio frequency communications can be implemented using a carrier signal or carrier wave modulated by a modulation signal, message signal and/or information signal that modulates the carrier wave. Carrier signals use periodic waves, repeating waveforms, and/or other predictable waveforms such as sinusoidal, square-waves, sawtooth, or other repeatable carriers which are then modulated in various ways by the message signal, modulation signal, and/or information signal.

(7) Therefore, it would be desirable to have methods, systems, and apparatuses that take into account at least some of the issues discussed above, as well as other possible issues. For example, it would be desirable to have methods and apparatus that overcome a technical problem with radio frequency communications using physical antennas. It would also be desirable to have systems, methods, and apparatuses that overcome the limitations of periodic and/or predictable carriers.

SUMMARY

(8) An embodiment of the present disclosure provides a communications system comprising a communications manager configured to identify data for transmission. The communications manager is configured to transmit pulses of noise signals encoding the data.

(9) Another embodiment of the present disclosure provides a communications system comprising a laser generation system, a computer system, and a communications manager. The laser generation system is configured to emit a set of laser beams; The communications manager is located in the computer system and the communications manager is configured to identify data for transmission. The communications manager is configured to control an emission of a set of laser beams to cause optical breakdown that generate pulses of radio frequency noise signals. The data is encoded in the pulses of the radio frequency noise signals.

(10) Yet another embodiment of the present disclosure provides a method for communicating data. Data is identified for transmission. Pulses of noise signals encoding the data are transmitted.

(11) Still another embodiment of the present disclosure provides a method for communicating data. Data for transmission is identified. Emission of a set of laser beams are controlled to cause optical breakdown that generate pulses of radio frequency noise signals. The data is encoded in the pulses of the radio frequency noise signals.

(12) Another embodiment of the present disclosure provides a method for communicating data. Pulses of noise signals are received. Data is encoded in the pulses of the noise signals. The data encoded in the pulses of the noise signals is decoded using a set of characteristics of the pulses of the noise signals.

(13) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1A is a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas in which illustrative embodiments may be implemented;
- (3) FIG. 1B is a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas from space in which illustrative embodiments may be implemented;
- (4) FIG. 2 is an illustration of a block diagram of a radio frequency communications environment in accordance with an illustrative embodiment;
- (5) FIG. 3 is an illustration of radio frequency noise generation using a laser beam in accordance with an illustrative embodiment;
- (6) FIG. 4 is an illustration of radio frequency noise generation using a plurality of laser beams in accordance with an illustrative embodiment;
- (7) FIG. 5 is an illustration of radio frequency noise generation using a plurality of laser beams in accordance with an illustrative embodiment;
- (8) FIG. 6 is an illustration of a diagram for controlling radio frequency noise generation in accordance with an illustrative embodiment;
- (9) FIG. 7 is an illustration of a diagram for controlling radio frequency noise generation in accordance with an illustrative embodiment;
- (10) FIG. 8 is an illustration of data transmission using pulse code noise modulation or pulse noise modulation in accordance with an illustrative embodiment;
- (11) FIG. 9 is an illustration of a block diagram of a communication system in accordance with an illustrative embodiment;
- (12) FIG. 10 is an illustration of a transmitter in accordance with an illustrative embodiment;
- (13) FIG. 11 is an illustration of a block diagram of a radio frequency transmitter in accordance with an illustrative embodiment;
- (14) FIG. 12 is an illustration of a block diagram of a receiver in accordance with an illustrative embodiment;
- (15) FIG. 13 is an illustration of a communications system for transmitting and receiving electromagnetic noise signals in accordance with an illustrative embodiment;
- (16) FIG. 14 is an illustration of a block diagram of a communications system for transmitting and receiving electromagnetic noise signals in accordance with an illustrative embodiment;
- (17) FIG. 15A is an illustration of a data flow of signals transmitting data using modulated noise signals in accordance with an illustrative embodiment;
- (18) FIG. 15B is an illustration of an envelope follower circuit using a diode detector with a low pass filter in accordance with an illustrative embodiment;
- (19) FIG. 15C is an illustration of a clipper circuit in accordance with an illustrative embodiment;
- (20) FIG. 16 is an illustration of a flowchart of a process for transmitting data in accordance with an illustrative embodiment;
- (21) FIG. 17 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;
- (22) FIG. 18 is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;
- (23) FIG. 19 is an illustration of a flowchart for controlling the emission of laser beams in points in

accordance with an illustrative embodiment

(24) FIG. **20** is an illustration of a flowchart of a process for transmitting data in accordance with an illustrative embodiment

(25) FIG. **21** is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

(26) FIG. **22** is an illustration of a flowchart for controlling the emission of the laser beams in points in accordance with an illustrative embodiment;

(27) FIG. **23** is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

(28) FIG. **24** is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

(29) FIG. **25** is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

(30) FIG. **26** is an illustration of a flowchart for controlling the emission of laser beams in points in accordance with an illustrative embodiment;

(31) FIG. **27** is an illustration a flowchart for controlling laser beams in accordance with an illustrative embodiment;

(32) FIG. **28** is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment;

(33) FIG. **29** is an illustration of a flowchart for transmitting pulses of noise signals in accordance with an illustrative embodiment;

(34) FIG. **30** is another illustration of a flowchart for transmitting pulses of noise signals in accordance with an illustrative embodiment;

(35) FIG. **31** is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment;

(36) FIG. **32** is an illustration of a flowchart of a process for controlling the emission of a set of laser beams in accordance with an illustrative embodiment;

(37) FIG. **33** is an illustration of a flowchart of a process for controlling the emission of a set of laser beams in accordance with an illustrative embodiment;

(38) FIG. **34** is an illustration of a flowchart for communicating data in accordance with an illustrative embodiment; and

(39) FIG. **35** is an illustration of a flowchart for decoding data in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

(40) The illustrative embodiments recognize and take into account one or more different considerations as described herein. For example, currently used physical antennas for transmitting radio frequency signals are subject to damage or destruction from various causes. For example, adverse weather conditions such as a hurricane or tornado can damage or destroy antennas such as transmission towers for land based communications. As another example, these physical antennas are also subject to damage or destruction from kinetic attacks.

(41) Thus, the illustrative embodiments provide a method, apparatus, system, and computer program product for transmitting of radio frequency signals without hardware such as transmission towers and physical antenna structures. In one or more illustrative examples we provide a non-physical radio frequency antenna that is impervious to adverse environmental conditions and kinetic attack. Further, these non-physical radio frequency antennas can be more difficult to detect.

(42) These transmitters can be positioned away from airplanes, transport, installations, buildings, or other locations that are subject to attack or undesired environmental conditions.

(43) In the illustrative examples, radio frequency transmissions transmitted using laser beams that induce or cause optical breakdowns to generate the radio frequency transmissions. In this illustrative example, the optical breakdowns create plasma that generates the radio frequency

signals. These optical breakdown points are the points of origination for transmitting the radio frequency signals.

(44) With reference now to the figures and, in particular, with reference to FIG. 1A, a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas is depicted in which illustrative embodiments may be implemented. As depicted, radio frequency signals can be transmitted from various platforms as depicted in this figure.

(45) As depicted, ground station **102** can transmit radio frequency signals **104** without using a physical antenna. In a similar fashion, ground station **106** can also transmit radio frequency signals **108** without using a physical antenna.

(46) In this example, laser beams are used by these ground stations to transmit the radio frequency signals. For example, ground station **102** emits laser beam **110** in a manner that causes optical breakdown **112** at optical breakdown point **114**. Radio frequency signals **104** are generated at and transmitted from optical breakdown point **114**.

(47) In this example, ground station **106** emits laser beam **116** and laser beam **118** at optical breakdown point **120** to cause optical breakdown **122**. In this example, two laser beams are used to cause optical breakdown **122** that results in transmission of radio frequency signals **108**.

(48) This type of transmission can be used from other platforms such as train **130**. In this example, train **130** emits laser beam **132** and laser beam **134** from different locations on train **130** at optical breakdown point **136**. The intersection of these two laser beams at optical breakdown point **136** causes optical breakdown **138**. As a result, radio frequency signals **142** are transmitted in response to optical breakdown **138** at optical breakdown point **136**.

(49) As another example, airplane **140** transmits radio frequency signals **142** using laser beam **144**. As depicted, laser beam **144** is emitted from airplane **140** at optical breakdown point **146**. Optical breakdown **148** occurs at optical breakdown point **146** which results in the transmission of radio frequency signals **142**.

(50) Turning now to FIG. 1B, a pictorial representation of platforms that can transmit radio frequency signals using non-physical antennas from space in which illustrative embodiments may be implemented. As another example, in FIG. 1B, satellite **160** emits laser **164** from space into the atmosphere **180** above earth **190** while satellite **162** emits laser **166** from space into the atmosphere **180** such that laser **164** and laser **166** intersect at optical breakdown point **168** causing optical breakdown **170** which results in radio frequency signals **172** originating and emanating from optical breakdown point **168**.

(51) As depicted, these radio frequency signals are generated without using physical antennas to transmit signals. Further, these radio frequency signals are transmitted at locations away from the platforms. As a result, identifying the platforms generating these radio frequency signals can be more difficult because antennas for transmitting the radio frequency signals are not visible. Further, tracking the location of where the radio frequency signals are generated does not provide identification of the platform or the platform location, nor the location of the communications system, computer system, communications manager, or the laser origination points in these examples.

(52) The locations of these optical breakdowns are considered radio frequency source emitters that can be in remote locations from the platforms emitting the laser beams. As a result, identifying the locations of the platforms becomes more difficult with the absence of physical antennas.

(53) Illustration of the different platforms in radio frequency communications environment **100** are only provided as examples of platforms that can implement this type of radio frequency signal transmission. In other illustrative examples, other platforms in addition to or in place of these platforms can be used. For example, this type of radio frequency generation can be implemented in a surface ship, a tank, or some other suitable type of platform in other illustrative examples.

(54) With reference now to FIG. 2, an illustration of a block diagram of a radio frequency communications environment is depicted in accordance with an illustrative embodiment. In this

illustrative example, radio frequency communications system **200** in communications environment **202** can communicate data **203** by using radio frequency signals **204** in the form of radio frequency noise signals **206**.

(55) Data **203** can take a number of different forms. For example, data **203** can be a document, a spreadsheet, sensor data, an image, a video, and email message, a text message, a webpage, a table, a data structure, serial data, commands, or other types of data that is to be transmitted or communicated. Data can also be analog or digital data. Analog and digital data can include, for example, music and audio. **

(56) In one illustrative example, a noise signal is a signal with irregular fluctuations that are or appear to be at least one of random, non-predictable, or non-deterministic.

(57) Further, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(58) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(59) A noise signal can be a signal that is statistically random. For example, a noise signal in these examples can be a signal that meets one or more standard tests for statistical randomness. A pseudorandom noise signal that seems to lack any definite pattern, although consisting of a deterministic sequence of pulses that repeats itself after its period is an example of a signal that is statistically random and considered a noise signal that can be used to encode data.

(60) In this example, radio frequency noise signals **206** are electromagnetic noise signals that can have a frequency from around 20 kHz to above the Terahertz range. Radio frequency noise signals can include signals with frequencies such as extremely low frequency (ELF), high frequency (HF), and other types of frequencies. These noise signals can also include microwave noise signals and Terahertz noise signals. Electromagnetic noise signals can also be optical noise in the visible range, infrared, ultraviolet X-rays and other types of noise signals that can be used as modulated noise. For example, lasers used at optical breakdown also may transmit as various ranges of noisy light in addition to noisy broadband radio frequencies. Modulating this noisy light with different techniques such as pulse noise modulation is included in this disclosure.

(61) In this illustrative example, radio frequency communications system **200** is associated with platform **208**. Platform **208** is an object that can transmit radio frequency noise signals **206** using radio frequency communications system **200**.

(62) Platform **208** can take a number of different forms. For example, platform **208** can be one of a mobile platform, a stationary platform, a land-based structure, an aquatic-based structure, and a space-based structure. More specifically, the platform can be a surface ship, a tank, a personnel carrier, a train, an airplane, a commercial airplane, a spacecraft, a space station, a satellite, a submarine, an automobile, a ground station, a power plant, a bridge, a dam, a house, a manufacturing facility, a building, and other suitable platforms.

(63) In this illustrative example, radio frequency communications system **200** comprises computer system **210** and communications manager **212**. In this example, communications manager **212** is located in computer system **210**.

(64) As used herein, “a number of” when used with reference to items, means one or more items. For example, “a number of different forms” is one or more different forms.

(65) Communications manager **212** can be implemented in software, hardware, firmware or a combination thereof. When software is used, the operations performed by communications

manager **212** can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by communications manager **212** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit.

(66) When hardware is employed, the hardware may include circuits that operate to perform the operations in communications manager **212**. The circuits used to implement communications manager **212** can take other forms in addition to or in place of a processor unit.

(67) In the illustrative examples, the hardware used to implement communications manager **212** can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform a number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

(68) Computer system **210** is a physical hardware system and includes one or more data processing systems. In this illustrative example, the data processing systems are hardware machines that can be configured to perform a sequence of operations. These operations can be performed in response to receiving an input in generating and output based on performing the operations. This output can be data in the form of values, commands, or other types of data. When more than one data processing system is present in computer system **210**, those data processing systems are in communication with each other using a communications medium. The communications medium may be a network. The data processing systems may be selected from at least one of a computer, a server computer, a tablet, or some other suitable data processing system.

(69) As depicted, computer system **210** includes a number of processor units **214** that are capable of executing program instructions **216** implementing processes in the illustrative examples. In other words, program instructions **216** are computer readable program instructions.

(70) As used herein, a processor unit in the number of processor units **214** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond and process instructions and program code that operate a computer. When the number of processor units **214** executes program instructions **216** for a process, the number of processor units **214** can be one or more processor units that are on the same computer or on different computers. In other words, the process can be distributed between processor units **214** on the same or different computers in a computer system **210**.

(71) Further, the number of processor units **214** can be of the same type or different type of processor units. For example, a number of processor units **214** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

(72) As depicted, radio frequency communications system **200** can also include laser generation system **218**. In other examples, laser generation system **218** can be considered a separate component controlled by radio frequency communications system **200**.

(73) In this example, laser generation system **218** is a hardware system that can emit a set of laser beams **220**. The operation of laser generation system **218** can be controlled by communications manager **212**.

(74) In this example, the set of laser beams **220** can be emitted from different locations **221**. For example, laser generation system **218** can be comprised of laser units that are positioned in

different locations. Each location can have one or more laser units for laser generation system **218** in this illustrative example.

(75) Communications manager **212** can identify data **203** for transmission using radio frequency noise signals **206**. Communications manager **212** controls an emission of a set of laser beams **220**. In this example, communications manager **212** directs or steers the set of laser beams **220** at a set of optical breakdown points **222**. In this example, the set of optical breakdown points **222** can be selected from at least one of intersection point **223** or focal point **225**

(76) As used herein, “a set of” when used with reference to items, means one or more items. For example, “a set of optical breakdown points **222**” is one or more of optical breakdown points **222**. In another example, a “set of laser beams” means one or more laser beams.

(77) In this example, intersection point **223** can be a location where two or more laser beams intersect. This location can be where an optical breakdown occurs from the intersection of two or more laser beams when the power **227** of two or more intersecting laser beams is sufficient to cause an optical breakdown. Focal point **225** can be a location where the laser beam is focused to cause an optical breakdown to occur at that location.

(78) This emission of the set of laser beams **220** is controlled by communications manager **212** to cause optical breakdowns **224** at the set of optical breakdown points **222** that generate radio frequency noise signals **206** encoding data **203**. In this illustrative example, plasma **226** occurs at optical breakdown points **222** in response to optical breakdowns **224** by the set of laser beams **220**. This plasma generated by optical breakdowns **224** causes radio frequency noise signals **206** to be transmitted at the set of optical breakdown points **222**.

(79) In this example, power **227** of laser beam **228** in the set of laser beams **220** at optical breakdown point **230** in the set of optical breakdown points **222** can be controlled using different mechanisms. For example, power **227** can be controlled by at least one of a shutter, a lens, a deformable lens, a microelectromechanical systems mirror, an attenuator, a controlling optics, an optical filter, an amplitude modulator in a laser beam generator, or other device.

(80) In this illustrative example, communications manager **212** can control the emission of the set of laser beams **220** by laser generation system **218** in a number of different ways. For example, communications manager **212** can control laser generation system **218** to emit a first number of the set of laser beams **220** continuously at the set of optical breakdown points **222**. Communications manager **212** can control laser generation system **218** to pulse a second number of the set of laser beams **220** at the set of optical breakdown points **222** to cause optical breakdowns **224** that generate radio frequency noise signals **206** encoding data **203**. The laser beam can be pulsed by turning the laser beam on and off. In another example, a laser beam can be pulsed by varying the power of the laser beam. In other words, the power can be pulsed by increasing and decreasing the power of the laser beam.

(81) In another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at the set of optical breakdown points **222** causing optical breakdowns **224** that generate radio frequency noise signals **206** encoding data **203**.

(82) In this example, the emission of the set of laser beams **220** can be performed in a number of different ways. The set of laser beams can be emitted as at least one of as pulsed or continuous. For example, one laser beam can be continuous while another laser beam is pulsed. Further, the laser beams can be originated from different directions at the set of optical breakdown points **222**.

(83) The direction at which a laser beam is emitted can move or sweep back such that an optical breakdown point is included during the movement of the laser beam. In other words, during the sweeping of the laser beam the laser beam can intersect with another laser beam. The intersection of this laser beam with another laser beam emitted the optical breakdown point can cause the optical breakdown at that optical breakdown point.

(84) In another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at selected optical breakdown point **232** in the set of

optical breakdown points **222**. Communications manager **212** can select new optical breakdown point **234** in the set of optical breakdown points as the selected optical breakdown point. Communications manager **212** can repeat emitting the set of laser beams **220** and selecting the new optical breakdown point while generating radio frequency noise signals **206** encoding data **203**.

(85) In yet another illustrative example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** from different locations **221** at optical breakdown point **230**. In this example, a portion of the set of laser beams **220** intersect at optical breakdown point **230** such that power **227** from the portion of the set of laser beams **220** is sufficient to cause optical breakdowns **224** at intersection point **223** that generate radio frequency noise signals **206** encoding data **203**.

(86) As another example, communications manager **212** can control laser generation system **218** to emit the set of laser beams **220** at optical breakdown point **230**. In this example, optical breakdowns **224** occur in response to all of the set of laser beams **220** intersecting at optical breakdown point **230**.

(87) In controlling the emission of the set of laser beams **220**, communication manager **212** can change a set of laser beam parameters **240** for the set of laser beams **220** to encode data **203** into radio frequency noise signals **206**. In this example, changing the set of laser beam parameters **240** changes a set of radio frequency characteristics **242** for radio frequency noise signals **206**. The set radio frequency characteristics **242** for radio frequency noise signals **206** can be selected from at least one of a timing, an optical breakdown point, an amplitude of the radio frequency noise signals, a frequency band, a relative phase, or other characteristics for radio frequency noise signals **206**.

(88) In yet another illustrative example, communications manager **212** can control laser generation system **218** to emit a subset of the set of laser beams **220** at the set of optical breakdown points **222** to cause the optical breakdowns **224** that generates radio frequency noise signals **206** encoding data **203**. Communications manager **212** can select a new subset of the set of laser beams **220** as the subset of laser beams **220**.

(89) Communications manager **212** can repeat emitting of the subset of the set of laser beams **220** and selecting the new subset of the set of laser beams **220** while transmitting radio frequency noise signals **206** encoding the data **203**.

(90) Thus, one or more illustrative examples enable transmitting radio frequency noise signals using radio frequency source emitters that do not require physical structures. As a result, one or more illustrative examples can overcome an issue with the vulnerability present in using physical source emitters such as antennas. In the illustrative examples, the optical breakdown points for the optical breakdowns are radio frequency source emitters.

(91) Further, these radio frequency source emitters can be moved almost instantaneously to different locations by repositioning the laser beams such that the laser beams point at different optical breakdown points. Attacks at these radio frequency source locations are attacks at the optical breakdown points where the plasma is generated. As a result, kinetic attacks against these locations are useless because the laser modulation sources are remote from the locations of these radio frequency source emitters.

(92) The illustration of communications environment **202** to in FIG. 2 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(93) For example, although communications manager **212** is shown as being implemented using program instructions **216** run on a number of processor units **214** in computer system **210**, communications manager **212** can be implemented in other hardware instead of or in addition to

the number of processor units **214**. For example, computer system **210** may use other hardware in addition to or in place of the number of processor units **214**.

(94) For example, other types of hardware circuits capable of performing the operations for communications manager **212** can be used. This other hardware can be at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations.

(95) Turning next to FIG. **3**, an illustration of radio frequency noise generation using a laser beam is depicted in accordance with an illustrative embodiment. This depicted example in FIG. **3** illustrates how a single laser beam can be used to generate radio frequency signals.

(96) In this illustrative example, laser generation system **300** is an example of laser generation system **218** in FIG. **2**. Laser generation system **300** emits laser beam **302**.

(97) As depicted, laser generation system **300** comprises a number of different components. In this example, laser generation system **300** includes oscillator **303** and optical system **304**.

(98) Oscillator **303** generates coherent light for emitting laser beam **302**. In this example, optical system **304** can focus laser beam **302**. Optical system **304** includes at least one of a lens, a mirror, or other optical element that can change the focus of laser beam **302**.

(99) In this example, the focus of laser beam **302** is controlled such that the power at focal point **306** is an optical breakdown point **308** where optical breakdown **310** occurs. As depicted in this example, optical breakdown **310** results in the generation of plasma **312**. Plasma **312** resulting from optical breakdown **310** causes the generation of radio frequency noise signal **314**. Thus, this example illustrates how a single laser beam can be used to generate radio frequency signals.

(100) Turning next to FIG. **4**, an illustration of radio frequency noise generation using a plurality of laser beams is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system **420** comprises laser unit **400** and laser unit **402**. Laser generation system **420** is an example of an implementation for laser generation system **218** in FIG. **2**.

(101) In this illustrative example, laser unit **400** generates first laser beam **404**. Laser unit **402** generates second laser beam **406**.

(102) In this example, first laser beam **404** and second laser beam **406** are emitted in directions from these laser beam units to intersect at optical breakdown point **408**. These two laser beams are emitted along different paths that intersect at optical breakdown point **408**. This optical breakdown point where the two laser beams intersect each other is intersection point **410**.

(103) In this example, the intersection of first laser beam **404** and second laser beam **406** results in optical breakdown **412**. This optical breakdown generates plasma **414**. As depicted in this example, optical breakdown **412** results in radio frequency noise signals **416**.

(104) As depicted in the example, optical breakdown **412** occurs where first laser beam **404** and second laser beam **406** intersect at intersection point **410**. In this example, the power for first laser beam **404** and second laser beam **406** individually is not sufficient to cause an optical breakdown.

(105) The illustration of the two laser units for laser generation system **420** in FIG. **4** is provided as an example of one implementation for generating radio frequency noise signals. This illustration is not meant to limit the manner in which other illustrative examples can be implemented. In another example, one or more laser units in addition to laser unit **400** and laser unit **402** can be used to generate additional laser beams. The laser beams can also intersect at intersection point **410** to cause optical breakdown **412**.

(106) Turning next to FIG. **5**, an illustration of radio frequency noise generation using a plurality of laser beams is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system **520** comprises laser unit **500** and optical system **502**. Laser generation system **520** is an example of an implementation for laser generation system **218** in FIG. **2**.

(107) In this example, laser unit **500** emits first laser beam **504** and second laser beam **506**. In this example, laser unit **500** generates initial laser beam **501** that is split into two laser beams, first laser

beam **504** and second laser beam **506** by optical system **502**.

(108) As depicted, optical system **502** comprises a number of different components. In this depicted example, optical system **502** comprises shutter **503**, variable attenuator **505**, beam splitter **507**, mirror **509**, and lens **511**.

(109) The components depicted are example components that can be used in optical system **502** and can change in other illustrative examples. For example, one or more of lens **511**, variable attenuator **505**, and shutter **503** may be omitted in other illustrative examples. In yet other illustrative examples, other components may be added such as a lens located before beam splitter **507**.

(110) As depicted, initial laser beam **501** is split into two laser beams by beam splitter **507**. Mirror **509** can be used to direct second laser beam **506** in different directions. Further, mirror **509** can be used to provide focus to increase the power of second laser beam **506** at a focal point such as optical breakdown point **508**. Lens **511** also can be used to provide focus to increase the power of second laser beam **506** at optical breakdown point **508**.

(111) In this example, first laser beam **504** and second laser beam **506** are emitted in directions to intersect at optical breakdown point **508**, which is intersection point **510** in this example. Optical breakdown **512** occurs at this intersection of first laser beam **504** and second laser beam **506**, generating plasma **514** that results in the generation of radio frequency noise signals **516**.

(112) In this example, the power of first laser beam **504** and second laser beam **506** are sufficient to cause optical breakdown **512** at the intersection of the laser beams. Optical breakdowns do not occur in other locations where these laser beams do not intersect each other in this example.

(113) With reference now to FIG. 6, an illustration of a diagram for controlling radio frequency noise generation is depicted in accordance with an illustrative embodiment. In this illustrative example, the operation of laser generation system **620** is controlled by controller **644**. As depicted, laser generation system comprises laser unit **600**, laser unit **602**, first power source **640**, second power source **642**, and optical system **605**. In this example, controller **644** can control the operation of laser generation system **620**.

(114) In this example, laser generation system **620** is an example of an implementation for laser generation system **218** in FIG. 2. Controller **644** is an example a component that can be implemented in communications manager **212** in FIG. 2.

(115) In this illustrative example, controller **644** can control the emission of first laser beam **604** and second laser beam **606** from laser generation system **620**. In this illustrative example, laser unit **600** generates first laser beam **604** using power supplied by first power source **640**. Laser unit **602** generates second laser beam **606** using power supplied by second power source **642**.

(116) In this example, first laser beam **604** and second laser beam **606** are emitted in directions that have paths that intersect at optical breakdown point **608**, which is intersection point **610**. Optical breakdown **612** occurs at this intersection of first laser beam **604** and second laser beam **606**, generating plasma **614** that results in the transmission of radio frequency noise signals **616**.

(117) In this example, controller **644** can control the emission of these laser beams such that at least one of first laser beam **604** or second laser beam **606** is pulsed. This pulsing can include at least one of turning a laser beam on and off for increasing and decreasing the power of the laser beam. This pulsing of one or both of first laser beam **604** and second laser beam **606** can be controlled to control the timing of radio frequency noise generation.

(118) When pulsed, optical breakdown **612** occurs when both laser beams intersect at intersection point **610**. When one laser beam is turned off, and intersection is not present between both laser beams and optical breakdown **612** does not occur. By controlling the timing of when first laser beam **604** and second laser beam **606** intersect at intersection point **610**, controller **644** can control the generation of radio frequency noise signals in a manner that encodes data.

(119) For example, data can be encoded in radio frequency noise signals based on the timing of when radio frequency noise signals are generated. As another example, the timing of the laser

beams can be used to control the duration of radio frequency noise signals. This duration can also be used to encode data into the radio frequency noise signals.

(120) In this illustrative example, controller **644** can control whether a laser unit emits a continuous laser beam or a pulsed laser beam using components such as first power source **640** and second power source **642**. These power sources can be turned on and off to turn the laser beams on and off. With this pulsing, optical breakdowns occur when both laser beams are on and intersect at intersection point **610**.

(121) In this example, the pulsing can also include increasing and decreasing the power in one or both of first laser beam **604** and second laser beam **606**. In this example, decreasing the power of one or both laser beams can prevent the occurrence of an optical breakdown because of insufficient power being present when first laser beam **604** and second laser beam **606** intersect at intersection point **610**. Optical breakdown **612** occurs when the power present from both laser beams intersecting at intersection point **610** is high enough for an optical breakdown.

(122) As another example, the pulsing of the laser beams can also be controlled using optical elements in optical system **605**. These optical elements can be controlled by controller **644** to pulse one or more of first laser beam **604** and second laser beam **606**.

(123) For example, variable attenuator **611** and shutter **613** can be operated to pulse first laser beam **604**. For example, shutter **613** can be used to selectively emit first laser beam **604**. Variable attenuator **611** can be used to change the power of first laser beam **604**. In similar fashion, the emission of second laser beam **606** can also be pulsed using variable attenuator **615** and shutter **617**.

(124) Thus, the emission of first laser beam **604** and second laser beam **606** from laser generation system **620** can be controlled by controller **644** such that both laser beams are continuous, one laser beam is continuous while the other laser beam is pulsed, or both laser beams are pulsed. This control can be performed to achieve optical breakdowns to transmit radio frequency noise signals in a manner that encodes data into the radio frequency signals.

(125) The illustration of laser generation system **620** is an example of one implementation and is not meant to limit the manner in which other illustrative examples can be implemented. For example, in other illustrative examples one or more laser units can be present in addition to laser unit **600** and laser unit **602**.

(126) With reference next to FIG. 7, an illustration of a diagram for controlling radio frequency noise generation is depicted in accordance with an illustrative embodiment. In this illustrative example, laser generation system **720** is controlled by controller **744**. As depicted, laser generation system **720** comprises laser unit **700**, power source **740**, and optical system **702**.

(127) Laser generation system **720** is an example of an implementation for laser generation system **218** in FIG. 2. Controller **744** is an example of the components that can be implemented in communications manager **212** in FIG. 2.

(128) In this illustrative example, controller **744** controls the emission of first laser beam **704** and second laser beam **706** from laser generation system **720**. In this illustrative example, laser unit **700** generates first laser beam **704** and second laser beam **706** using power supplied by power source **740**. In this example, laser unit **700** generates initial laser beam **701** that is split into two laser beams, first laser beam **704** and second laser beam **706** by optical system **702**.

(129) As depicted, optical system **702** comprises a number of different components. In this example, optical system **702** comprises shutter **703**, variable attenuator **705**, beam splitter **707**, and mirror **709**, mirror **711**, mirror **713**, and lens **715** as other components that can be located in optical system **702**. The components depicted are example components that can be used in optical system **702** and these components can change in other illustrative examples. For example, one or more lens **715**, variable attenuator **705**, and shutter **703** may be omitted in other illustrative examples. In yet other illustrative examples, other components may be included such as a lens located before beam splitter **707**.

(130) As depicted, initial laser beam **701** is split into two laser beams by beam splitter **707**. In this example, first laser beam **704** and second laser beam **706** are emitted in directions to intersect at optical breakdown point **708**, which is intersection point **710** in this example. Optical breakdown **712** occurs at this intersection of first laser beam **704** and second laser beam **706**. Optical breakdown **712** generates radio frequency signal **716** through plasma **714** occurring from optical breakdown **712**.

(131) In this example, the power of the first laser beam **704** and second laser beam **706** are sufficient to cause optical breakdown **712** at the intersection of the laser beams. Optical breakdowns do not occur in other locations where these laser beams do not intersect each other in this example.

(132) In this example, controller **744** can control the emission of these laser beams such that at least one of first laser beam **704** or second laser beam **706** is pulsed. This pulsing can include at least one of turning a laser beam on and off for increasing and decreasing the power of the laser beam. This pulsing of one or both of first laser beam **704** and second laser beam **706** can be controlled to control the timing of radio frequency noise generation.

(133) In this example, first laser beam **704** can be pulsed by controller **744** controlling the operation of at least one of variable attenuator **705** or shutter **703**. Variable attenuator **705** can be used to change the power of first laser beam **704**. Shutter **703** can turn laser beam on and off with respect to emissions of laser beams from laser generation system **720**. In this example, both laser beams can be pulsed at the same time by controlling power source **740**. In another illustrative example, components within laser unit **700** such as an amplitude modulator can be controlled to pulse the power of initial laser beam **701** resulting in a pulsing of both first laser beam **704** and second laser beam **706**.

(134) By controlling the timing of when first laser beam **704** and second laser beam **706** intersect at intersection point **710**, controller **744** can control the generation of radio frequency noise signals in a manner that encodes data.

(135) In yet another illustrative example, controller **744** can control the location of optical breakdown point **708** by moving one or both of first laser beam **704** and second laser beam **706**. This movement of optical breakdown point **708** can be controlled using at least one of mirror **709** or mirror **713**. By moving the location of optical breakdown point **708**, the phase of radio frequency noise signal can be changed to encode data.

(136) The illustration of example implementations for laser generation system **218** in FIG. 2 and in FIGS. 3-7 have been provided as an example of some illustrative examples and are not meant to limit the manner in which other laser generation systems can be implemented. For example, a laser generation system can include both a first laser unit and a second laser unit with an optical system. In yet another illustrative example, one or more laser units can be present that emit laser beams in addition to the ones depicted at different optical breakdown points. With this example, two or more optical breakdowns can occur from laser beams emitted from a laser generation system.

(137) In yet another illustrative example, different laser beams can be emitted at different times at the same optical breakdown point. As a result, optical breakdowns can be generated from different combinations of laser beams at the same optical breakdown point.

(138) The illustrative embodiments also recognize and take into account that current techniques for transmitting data involves the use of carrier wave forms. For example, many techniques use only periodic, sinusoidal, or other repetitive or predictable carrier wave forms that are modulated to encode data. These types of waveforms can be detected in noise through various techniques including the denoiser technology which can detect sinusoidal carriers at 20 dB to 40 dB below a noise floor.

(139) As a result, interception and decoding of signals can occur using current transmission techniques. Further, when the sinusoidal carriers can be detected, security issues can arise. For example, information can be inserted into transmissions, jamming attacks can occur, or other issues

with using single sinusoidal, periodic, or other repetitive carriers to transmit data.

(140) Thus, the illustrative embodiments provide a method, apparatus, and system for transmitting data. In the illustrative examples, this data can be transmitted using various modulation techniques that modulate noise signals. The use of noise signals is in contrast to the use of a sinusoidal, periodic, repetitive, or predictable carrier that can be detected.

(141) Turning to FIG. 8, an illustration of data transmission using pulse code noise modulation (also called pulse noise modulation) is depicted in accordance with an illustrative embodiment. In this illustrative example, pulse code noise modulation or pulse noise modulation can be performed using a radio frequency communication system such as radio frequency communication system 200 in FIG. 2.

(142) In this illustrative example, optical breakdowns are generated over time. These optical breakdowns result in the generation of plasma 800 that causes radio frequency noise signals 802 to be transmitted.

(143) The timing of these optical breakdowns can be selected to encode data such that the generation of radio frequency noise signals 802 encode the data. In this example, pulses are present in radio frequency noise signals 802 with timing that corresponds to the timing of optical breakdowns that generated plasma 800. These pulses of radio frequency noise signals 802 are timed to encode data. This type of encoding of data can be referred to as pulse noise modulation. As depicted, radio frequency noise signals 802 can be received and decoded to obtain decoded data signal 804.

(144) This illustration of using radio frequency noise signals generated by optical breakdowns to communicate data is presented as one example of how pulses of radio frequency noise signals can encode data. This illustration is not meant to limit the manner in which other illustrative examples can be implemented.

(145) For example, the pulses of radio frequency noise signals can be generated using other techniques in addition to or in place of laser-induced optical breakdowns. A transmitter system can use a noise signal as a carrier signal and a modulator to modulate the carrier signal such that pulses of radio frequency noise are transmitted that encode the data.

(146) In still other illustrative examples, other types of noise signals in addition to or in place of radio frequency electromagnetic noise signals can be used. For example, noise signals can be used for transmitting data encoded in pulses and can be selected from at least one of electromagnetic frequency noise signals, radio frequency noise signals, microwave frequency signals, audio frequency noise signals, ultrasonic frequency noise signals, ultra-low frequency noise signals, very low frequency noise signals, underwater frequency noise signals, optical frequency noise signals, or other types of noise signals. These different types of noise signals can be used for various applications including speech communication, music, or other types of information for data that that are encoded in the noise signals.

(147) With reference next to FIG. 9, an illustration of a block diagram of a communication system is depicted in accordance with an illustrative embodiment. In this illustrative example, communications system 900 in communications environment 901 operates to transmit data 902 encoded in noise signals 904.

(148) In one illustrative example, a noise signal is a signal with irregular fluctuations that are or appear to be random, non-predictable, or non-deterministic. A noise signal can be a signal that is statistically random. For example, a noise signal in these examples can be a signal that meets one or more standard tests for statistical randomness. A pseudorandom noise signal that seems to lack any definite pattern, although consisting of a deterministic sequence of pulses that repeats itself after its period is an example of a signal that is statistically random and considered a noise signal that can be used to encode data. In this example, the noise in noise signals 904 can be selected from at least one of nondeterministic noise, pseudo random noise, or some other suitable type of noise signal.

(149) In the illustrative example, signals can have characteristics selected from at least one of amplitude, frequency, bandwidth, timing, phase, or other characteristics. In this illustrative example, noise signals **904** can be noise signals in which at least one of these characteristics are not controlled to encode the data. In other words, at least one or more of these characteristics meet one or more standard tests for statistical randomness in noise signals **904**.

(150) In these examples, noise signals **904** do not include carrier waves that are periodic. These types of signals can be, for example, sinusoidal, sawtooth, square, or other types of signals. Noise signals **904** also do not include periodic or sinusoid-based carrier signals that employ spread spectrum, frequency-hopping signals, and radar “chirps” that are based on periodic signals such as sinusoids or sawtooths. These and other types of signals that do not meet one or more standard tests for statistical randomness are not considered noise signals **904** in this example. However, “spread noise spectrum”, frequency-hopping noise signals, and noise-based radar bursts that use noise as the basis of their carrier signals are considered noise signals **904** in this example.

(151) As depicted, communications system **900** comprises computer system **910** and communications manager **912** located in computer system **910**.

(152) Communications manager **912** can be implemented in software, hardware, firmware or a combination thereof. When software is used, the operations performed by communications manager **912** can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by communications manager **912** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit.

(153) When hardware is employed, the hardware may include circuits that operate to perform the operations in communications manager **912**.

(154) The circuits used to implement communications manager **912** can take other forms in addition to or in place of a processor unit.

(155) In the illustrative examples, the hardware used to implement communications manager **912** can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform the number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

(156) Computer system **910** is a physical hardware system and includes one or more data processing systems. In this illustrative example, the data processing systems are hardware machines that can be configured to perform a sequence of operations. These operations can be performed in response to receiving an input in generating and output based on performing the operations. This output can be data in the form of values, commands, or other types of data. When more than one data processing system is present in computer system **910**, those data processing systems are in communication with each other using a communications medium. The communications medium may be a network. The data processing systems may be selected from at least one of a computer, a server computer, a tablet, or some other suitable data processing system.

(157) As depicted, computer system **910** includes a number of processor units **914** that are capable of executing program instructions **916** implementing processes in the illustrative examples. In other words, program instructions **916** are computer readable program instructions.

(158) As used herein, a processor unit in the number of processor units **914** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond and

process instructions and program code that operate a computer. When the number of processor units **914** executes program instructions **916** for a process, the number of processor units **914** can be one or more processor units that are on the same computer or on different computers. In other words, the process can be distributed between processor units **914** on the same or different computers in a computer system **910**.

(159) Further, the number of processor units **914** can be of the same type or different type of processor units. For example, a number of processor units **914** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

(160) As depicted, communications system **900** can also include signal transmission system **918**. In other examples, signal transmission system **918** can be considered a separate component controlled by communications system **900**.

(161) In this depicted example, signal transmission system **918** is a hardware system that can transmit noise signals **904**. The operation of signal transmission system **918** can be controlled by communications manager **912**.

(162) In this illustrative example, noise signals **904** are received by receiver **920**. Receiver **920** is also depicted as part of communication system **900**. In yet other illustrative examples, receiver **920** may be a separate component from communications system **900**.

(163) Receiver **920** is a hardware system and can include processes implemented in hardware or software that decode data **902** that is encoded in pulses **922** of noise signals **904**.

(164) In this illustrative example, communications manager **912** identifies data **902** for transmission. In response to identifying data **902**, communications manager **912** transmits pulses **922** of noise signals **904** encoding data **902**. In one illustrative example, data **902** can be encoded in pulses **922** of noise signals **904** using at least one of a timing of the pulses **922**, an amplitude of the pulses **922**, duration of pulses **922**, or other characteristic for pulses **922**. In this manner, communications manager **912** can perform pulse noise modulation through the modulation of noise signals **904** to encode data **902**.

(165) For example, communications manager **912** can control the operation of signal transmission system **918** to perform pulse modulation **924**. With pulse modulation **924**, pulses **922** can encode data **902** through the timing of pulses **922** which are noise pulses or pulses of noise in this example.

(166) For example, the presence of a noise pulse or pulse of noise can be considered a “1” and the absence of a noise pulse or pulse of noise can be considered a “0” which can be selected in time to encode data **902**. The timing of the presence or absence of pulses **922** of noise can occur using various time periods.

(167) For example, the timing can be based on whether a noise pulse or pulse of noise is present or absent at each period of time. The period of time can be, for example, a microsecond, a millisecond, two milliseconds, or some other period of time during which a pulse is absent or present for encoding data **902** in pulses **922** of noise.

(168) With reference next to FIG. **10**, an illustration of a transmitter is depicted in accordance with an illustrative embodiment. In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures. In this illustrative example, examples of components that can be used to implement signal transmission system **918** in FIG. **9** are depicted.

(169) As depicted in this illustrative example, signal transmission system **918** can include a number of different components that can be controlled to transmit noise signals **904**. More specifically, these components can be controlled to generate pulses **922** of noise signals **904**. These components can include at least one of laser generation system **1000** or radio frequency transmitter **1002**.

(170) In this illustrative example, laser generation system **1000** is a hardware system that emits a set of laser beams **1004**. Communications manager **912** can control the emission of the set of laser

beams **1004** from laser generation system **1000** to cause optical breakdowns **1005**.

(171) In this example, optical breakdowns **1005** result in the generation of noise signals **904** in the form of radio frequency noise signals **1006**. In this example, pulses **922** of radio frequency noise signals **1006** can be generated based on the timing of optical breakdowns **1005**. In this illustrative example, each optical breakdown in optical breakdowns **1005** can be a pulse in pulses **922** of radio frequency noise signals **1006**.

(172) In this example, radio frequency transmitter **1002** is a hardware system and can transmit pulses **922** of noise signals **904** in the form of radio frequency noise signals **1006**. For example, radio frequency transmitter **1002** can transmit pulses **922** of noise signals **904** in the form of radio frequency noise signals **1006** transmitted from a physical hardware antenna instead of using lasers and optical breakdowns to produce the radio frequency noise signals **1006**.

(173) Turning next to FIG. **11**, an illustration of a block diagram of a radio frequency transmitter is depicted in accordance with an illustrative embodiment. This figure illustrates example components that can be used to implement radio frequency transmitter **1002**. As depicted in this example, radio frequency transmitter **1002** comprises electrical noise generator **1100**, modulator **1102**, and transmitter **1104**.

(174) As depicted, electrical noise generator **1100** generates carrier noise signal **1106**. Electrical noise generator **1100** is connected to modulator **1102** and sends carrier noise signal **1106** to modulator **1102**.

(175) As depicted, modulator **1102** receives data **1108** that is to be transmitted. In this example, modulator **1102** modulates carrier noise signal **1106** to create pulsed carrier noise signal **1110** that encodes data **1108**. This data is encoded in pulses in pulsed carrier noise signal **1110**. In this example the modulation occurs by modulator **1102** turning carrier noise signal **1106** on and off to form pulsed carrier noise signal **1110**.

(176) Transmitter **1104** transmits pulsed carrier noise signal **1110** as pulses **1112** of radio frequency noise signals **1114**. In this example, transmitter **1104** includes a physical antenna that is used to transmit pulses **1112** of radio frequency noise signals **1114**. In other illustrative examples, the antenna can be a separate component from the hardware used to generate radio frequency noise signals **1114**.

(177) Turning next to FIG. **12**, an illustration of a block diagram of a receiver is depicted in accordance with an illustrative embodiment. An example of components that can be used to implement receiver **920** are shown in this figure. As depicted, receiver **920** is a hardware system. As depicted, receiver **920** comprises broadband radio frequency receiver **1200**, frequency selector **1202**, and clipper circuit **1204**.

(178) In this illustrative example, broadband radio frequency receiver **1200** receives radio frequency noise signals **1206**. Broadband radio frequency receiver **1200** is connected to frequency selector **1202** and sends the received signals to frequency selector **1202**.

(179) Frequency selector **1202** outputs a set of voltage signals **1208** from the frequencies selected in radio frequency noise signal **1206**. In this illustrative example, the selection of frequencies by frequency selector **1202** can be performed using at least one of a bandpass filter, a band-reject filter, an envelope follower, an envelope detector, a low-pass filter, a rectified low pass filter, multiple bandpass filters tuned to different frequencies, or some other suitable type of circuit.

(180) Frequency selector **1202** is connected to clipper circuit **1204**. Voltage signal **1208** is received by clipper circuit **1204**, which shapes voltage signal **1208**. In this illustrative example, clipper circuit **1204** prevents voltage signal **1208** from exceeding a selected voltage level. Clipper circuit **1204** outputs data signal **1210**. In this example, data signal **1210** is in an analog or digital signal and contains pulses that can be used re-create the data transmitted in radio frequency noise signals **1206**.

(181) Thus, one or more illustrative examples enable communicating data using noise carrier signals. In one illustrative example, these noise carrier signals or carrier noise signals can be

modulated to encode data. The modulation can be pulse noise modulation or pulse code noise modulation in which a noise signal is transmitted in pulses. The timing of the pulses selected encodes data in these pulses of noise signals.

(182) In this illustrative example, the modulation and demodulation of these pulses of noise signals do not depend on a single frequency or periodic waveform as the basis for the carrier wave as compared to current techniques that use a sinusoidal, periodic, or predictable carrier. As result, increased security can be present and interference with the sinusoidal carriers can be reduced.

(183) In one illustrative example, the pulse code noise modulation or pulse noise modulation can be a broadband noise radio frequency carrier signal encoding the data. The generation of the pulses of radio frequency noise signals can be performed using a laser generation system that generates radio frequency signals through optical breakdowns. In another example, the generation of the radio frequency noise signals can be performed using a physical electromagnetic receipt transmitter having a physical antenna.

(184) The illustration of communications environment **901** and the different components in FIGS. **9-12** is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(185) For example, although communications manager **912** is shown as being implemented using program instructions **916** run on a number of processor units **914** in computer system **910**, communications manager **912** can be implemented in other hardware instead of or in addition to the number of processor units **914**. For example, computer system **910** may use other hardware in addition to or in place of the number of processor units **914**.

(186) For example, other types of hardware circuits capable of performing the operations for communications manager **912** can be used. This other hardware can be at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations.

(187) As another example, the illustration of laser generation system **1000** and radio frequency transmitter **1002** are provided as examples of some implementations of components that can transmit pulses **922** of noise signals **904**. As another example, radio frequency transmitter **1002** that generates electrical noise in electrical noise generator **1100** as carrier signal noise **1106** to be modulated and transmitted as radio frequency noise signals **1114** in noise pulses **1112** can be transmitted on any type of physical, hardware antenna, or both. Examples of antenna types include, for example, whip antennas, dipole antennas, microwave antennas, metamaterial antennas, directional antennas, omnidirectional antennas, and any other type of physical antenna.

(188) Turning next to FIG. **13**, an illustration of a communications system for transmitting and receiving electromagnetic noise signals is depicted in accordance with an illustrative embodiment. In this illustrative example, communications system **1300** can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals. Examples of electromagnetic noise signals may include electromagnetic ranges of ULF (Ultra Low Frequency), VLF (Very Low Frequency), 20 KHz to 300 KHz, HF (High Frequency), UHF (Ultra High Frequency), millimeter wave and microwave ranges, EHF (Extremely High Frequencies) up through Gigahertz frequencies and above Terahertz frequencies, and including the optical spectrum. Examples of applications or uses include noise carrier communications for modulation of audio, voice, and video communications, as well as noise-based radar, noise-based precision navigation and timing such as noise-based global positioning systems, noise-based spread spectrum using frequency bands of noise instead of sinusoidal-based carrier spread spectrum, noise-based frequency band-hopping using frequency-hopping of frequency bands of noise instead of using sinusoidal or periodic-based

carrier frequency-hopping, as well as Signals Intelligence (SI) waveforms such as Low Probability of Intercept/Low Probability of Detect (LPI/LPD) and other clandestine signaling where detection and interception of messages using noise carriers will be difficult.

(189) As depicted, modulator **1302** receives data signal **1304** and carrier noise signal **1306**. In this illustrative example, carrier noise signal **1306** can be generated by an electrical noise generator.

(190) Modulator **1302** modulates carrier noise signal **1306** to generate modulated signal **1308**. In this example, modulated signal **1308** comprises pulses of carrier noise signal **1306**. For example, modulated signal **1308** can be generated by turning modulator **1302** on and off to send pulses of carrier noise signal **1306** to transmitter **1310** for transmission as modulated signal **1308**. The generation of the pulses is based on the data in data signal **1304**. In this manner, the data in data signal **1304** can be encoded in modulated signal **1308**.

(191) Transmitter **1310** transmits modulated signal **1308** to receiver **1312**. In one illustrative example, receiver **1312** can be a broadband radio frequency receiver when modulated signal **1308** is a radio frequency signal. When other types of signals are used, receiver **1312** is selected to detect the signals transmitted by transmitter **1310**.

(192) Modulated signal **1308** detected by receiver **1312** is sent to demodulator **1314**. In this example, demodulator **1314** demodulates modulated signal **1308** using carrier noise signal **1320** to generate data signal **1318**, which contains the same data in data signal **1304** in this depicted example.

(193) As depicted, the demodulation of modulated signal **1308** is performed using carrier noise signal **1320**. In this illustrative example, carrier noise signal **1306** is not predictable as compared to current techniques using sinusoidal wave forms for carrier signals.

(194) As depicted, carrier noise signal **1320** can be obtained by demodulator **1314** in the form of unmodulated carrier noise signal **1322** being transmitted to demodulator **1314**. In this manner, carrier noise signal **1320** used to demodulate modulated signal **1308** can be the same carrier signal as carrier noise signal **1306**. Unmodulated carrier noise signal **1322** can be an in-band or out-of-band copy of carrier noise signal **1306**.

(195) With reference now to FIG. **14**, an illustration of a block diagram of a communications system for transmitting and receiving electromagnetic noise signals is depicted in accordance with an illustrative embodiment. Communications system **1400** can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals.

(196) In this illustrative example, communications system **1400** can transmit or receive electromagnetic noise signals encoding data using electromagnetic noise signals.

(197) As depicted, modulator **1402** receives data signal **1404** and carrier noise signal **1406**. In this illustrative example, carrier noise signal **1406** can be generated by an electrical noise generator.

(198) Modulator **1402** modulates carrier noise signal **1406** to generate modulated signal **1408**. In this example, modulator **1402** can be an on/off modulator. As an on/off modulator, modulator **1402** sends carrier noise signal **1406** transmitter **1410** for transmission when modulator **1402** is turned on and does not send carrier noise signal **1406** to transmitter **1410** when modulator **1402** is turned off. As result, modulated signal **1408** comprises pulses of carrier noise signal **1406**. These pulses are generated to encode data signal **1404**. In other words, the timing of these pulses can be generated to encode the data. For example, the timing in these depicted examples can be time for pulses to perform pulse code noise modulation or pulse noise modulation.

(199) For example, modulated signal **1408** can be generated by turning modulator **1402** on and off to send pulses of carrier noise signal **1406** to transmitter **1410** for transmission as modulated signal **1408**.

(200) Transmitter **1410** transmits modulated signal **1408** to receiver **1412**. In one illustrative example, receiver **1412** can be a broadband radio frequency receiver when modulated signal **1408** is a radio frequency signal. When other types of signals are used, receiver **1412** is selected to detect the signals transmitted by transmitter **1410**.

(201) In this example, modulated signal **1408** detected by receiver **1412** is sent to envelope follower **1414**. As depicted, envelope follower **1414** can also be referred to as an envelope detector. Envelope follower **1414** can detect amplitude variations in modulated signal **1408** and create a signal having a shape that resembles those variations. This example, modulated signal **1408** contains pulses of noise. As a result, envelope follower **1414** can generate a signal with the shape of the noise pulses to form data signal **1416**. Envelope follower **1414** can be a selected from at least one of a low pass filter, a bandpass filter, an envelope detector, a peak detector, or a diode detector that follows and outputs the overall shape of at least one of the amplitudes or pulses as currently used.

(202) The illustrative examples of communication systems in FIG. **13** and in FIG. **14** are presented as examples of some implementations for communications system **900** in FIG. **9**. These illustrations are not meant to limit the manner in which other illustrative examples can be implemented. For example, a clipper circuit as is known in the art can be placed after envelope follower **1414** in FIG. **14** to convert rough envelopes of pulses into square wave pulses.

(203) Turning now to FIG. **15A**, an illustration of a data flow of signals transmitting data using modulated noise signals is depicted in accordance with an illustrative embodiment. In this illustrative example, data signal **1501** is an example of signals in communications system **1400** in FIG. **14**.

(204) In this illustrative example, data signal **1500** is used to modulate carrier noise signal **1502**. Data signal **1500** is an example of data signal **1404** and carrier noise signal **1502** is an example of carrier noise signal **1406** in FIG. **14**.

(205) The modulation of carrier noise signal **1502** forms modulated signal **1504**, which encodes the data in data signal **1500**. Modulated signal **1504** is an example of modulated signal **1408** in FIG. **14**. As depicted in this example, modulated signal **1504** is a modulated noise signal comprising pulses of carrier noise signal **1502**.

(206) Received signal **1506** is an example of the signal received by a receiver. As depicted, received signal **1506** also includes noise **1508** in addition to the pulses of carrier noise signal **1502** in modulated signal **1504**. In this example, noise **1508** is background noise or other noise in addition to the pulses in the carrier noise in modulated signal **1504**.

(207) As depicted, received signal **1506** can be processed and decoded using a component such as envelope follower **1414** in FIG. **14**. Other components such as a bandpass filter, low-pass filter, band reject filter, clipper circuit, or other circuits can be used to generate data signal **1510**. In this example, data signal **1510** is the same as or close enough to data signal **1500** such that the same data used to generate data signal **1500** can be obtained from data signal **1510**.

(208) As discussed previously, the set of characteristics for noise signals can be selected from at least one of a timing, an amplitude, a frequency band, a relative phase, or other characteristics for carrier noise signals. For pulse noise modulation the carrier noise may be of different frequency characteristics that the transmitter and receiver will share. For pulse noise modulation the carrier noise signals will vary in amplitude, duration, and timing to modulate the message signal. For reception of these pulse noise modulated signals the receivers in FIG. **12** and FIG. **14** use various types of techniques to receive and demodulate the original data signal.

(209) Turning now to FIG. **15B**, an illustration of an envelope follower circuit using a diode detector with a low pass filter in accordance with an illustrative embodiment. In this figure, an illustration of a simple circuit for envelope follower **1414** is shown. In this illustrative example, envelope follower **1414** is comprised of a diode **1512** to rectify the input signal, capacitor **1514** to provide a low pass filter to smooth out the noisy rectified signal and produce a lower frequency envelope. An optional resistor **1516** or inductive coil may be provided to affect the tuning or resonance of the circuit. Here, received signal **1506** of carrier pulses of noise is inputted to the envelope follower **1414** circuit. As received signal **1506** travels through diode **1512** the diode acts as a rectifier and converts the AC noise signal into a DC noise signal **1518** as shown by the dashed

arrow from DC noise signal **1518** to the output of diode **1512**. From there the rectified DC noise signal **1518** travels across capacitor **1514** which acts as a low pass filter to smooth the signal into an envelope signal **1520**. The actual envelope signal **1520** is shown by the line the follows the outline or envelope of the noise bursts from DC noise signal **1518**. The envelope signal **1520** then travels across optional resistor **1516** or coil and exits at the output as the envelope followed signal **1522**.

(210) In this illustrative example it is clear that the envelope followed signal **1522** is beginning to look like the received data signal **1510**.

(211) With reference to FIG. **15C**, an illustration of a clipper circuit in accordance with an illustrative embodiment. In this figure, clipper circuit **1204** is also referred to as a slicer or amplitude selector. In this illustrative example, clipper circuit **1204** is comprised of optional input resistor **1524**, and a bidirectional clipping circuit comprised of diode D1 **1526**, bias voltage **1528**, diode D2 **1530**, and bias voltage **1532**. This and many other known methods of clipping can be used. Single directional clipping may be used as well as bidirectional or any other type of clipping circuit.

(212) In this illustrative example, envelope followed signal **1522** from FIG. **15B** has been amplified to be a stronger signal and is inputted into clipper circuit **1204**. Envelope followed signal **1522** travels through optional input resistor **1524** which may be an impedance matching circuit.

(213) This signal then travels across one or more illustrative diodes D1 **1526** and D2 **1530**. Various types of diodes may be used. A single diode may be used, or a transistor circuit may be used with the purpose of clipping off the top of envelope followed signal **1522** such that top part of signal **1534** is clipped off and bottom part of signal **1536** remains. The level at which top part of the signal **1534** is clipped off is determined by the diodes D1 **1526** and D2 **1530** as well as by the bias voltages **1528** and **1532**.

(214) Thus, bottom part of signal **1536** remaining is outputted at the output. This bottom part of the signal **1536** can be transferred through another stage of clipping until it becomes output data signal **1510** which is extremely similar to the original data signal **1500**.

(215) As can be seen in this illustrative example, the pulses of carrier noise signal **1502** encode data in data signal **1500**. In other words, the timing in generating pulses of carrier noise signal **1502** is used to encode the data.

(216) Thus, the different illustrative examples use pulse modulation of a noise signal that can be generated using a laser generator or a transmitter. With a laser generator, optical breakdowns are used to create the pulses of noise signals. With a physical transmitter, an electronic noise source generates a carrier noise signal that is modulated to create pulses of the carrier noise signal based on the data to be transmitted. These pulses of the carrier noise signals form the pulses of noise signal encoding data that can be transmitted using a physical antenna.

(217) In this illustrative example, FIGS. **16-27** are flowcharts illustrating operations that can be performed to generate radio frequency noise signals encoding data in which a physical antenna is unnecessary.

(218) With reference first to FIG. **16**, a flowchart of a process for transmitting data is depicted in accordance with an illustrative embodiment. The process in FIG. **16** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager **212** in computer system **210** in FIG. **2**.

(219) The process begins by identifying data for transmission (operation **1600**). The process controls an emission of a set of laser beams to cause optical breakdowns generating radio frequency noise signals encoding the data (operation **1602**). The process terminates thereafter.

(220) In operation **1602**, the emission of the set of laser beams can be controlled in number of different ways. For example, the laser beams can be emitted continuously or pulsed. Further,

direction at which the laser beams are directed can also be changed. For example, the set of laser beams can be directed towards a set of optical breakdown points. The optical breakdown points can be selected from at least one of an intersection point or focal point. These optical breakdown points are locations where optical breakdowns occur. These optical breakdowns are locations where plasma is generated that generates the radio frequency noise signals.

(221) The manner in which the optical breakdowns occur can be used to encode the data in the radio frequency noise signals. For example, the timing of the occurrence of optical breakdowns generate time pulses used to encode data. In this manner, different types of data encoding such as pulse noise modulation can be used to encode data based on when radio frequency noise signals are generated.

(222) As another example, the set of laser beams can be moved or swept such that the optical breakdowns occur in different locations resulting in the frequency of a phase change in the optical breakdowns that can be used to encode data. As another example, the power of the laser beams can be changed to change the amplitude of the radio frequency noise signals to encode data. In this manner, different types of data encoding such as pulse noise modulation can be used to encode data based on when radio frequency noise signals are generated.

(223) Turning next to FIG. 17, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 17 is an example of one implementation for operation 1602 in FIG. 16.

(224) The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1700). The process terminates thereafter.

(225) With reference next to FIG. 18, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 18 is another example of an implementation for operation 1602 in FIG. 16.

(226) The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1800). The process terminates thereafter.

(227) In FIG. 19, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 19 is yet another example of an implementation for operation 1602 in FIG. 16.

(228) The process controls emission of the set of laser beams to intersect an intersection point such that a power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the radio frequency noise signals encoding the data (operation 1900). The process terminates thereafter.

(229) Turning next to FIG. 20, a flowchart of a process for transmitting data is depicted in accordance with an illustrative embodiment. The process in FIG. 20 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager 212 in computer system 210 in FIG. 2.

(230) The process begins by identifying the data for transmission using radio frequency noise signals (operation 2000). The process controls an emission of laser beams at a set of optical breakdown points to cause optical breakdowns that generate the radio frequency noise signals encoding the data (operation 2002). The process terminates thereafter.

(231) In this example, the set of optical breakdown points can be at different locations when more than one optical breakdown point is present in the set of optical breakdown points. In one example, radio frequency transmissions can be transmitted from multiple locations when the set of optical breakdowns caused by the set of lasers being directed at more than one optical breakdown point.

(232) In FIG. 21, an illustration of a flowchart for controlling the emission of laser beams is

depicted in points with an illustrative embodiment. The process illustrated in FIG. 21 is yet another example of an implementation for operation **2002** in FIG. 20.

(233) The process begins by emitting a first set of the laser beams continuously at the set of optical breakdown points (operation **2100**). The process pulses a second set of the laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation **2102**). The process terminates thereafter.

(234) In operation **2102**, the pulsing can occur by turning the second set of laser beams on and off. In other examples, the pulsing can provide increasing decreasing the power to the second set of laser beams. In this example, the optical breakdowns occur in response to sufficient power in the laser beams at the set of optical breakdown points. In this example, the pulsing can control the timing of when radio frequency noise signals are transmitted.

(235) Further in operation **2102**, a power of a laser beam at the optical breakdown point can be controlled at by at least one of a shutter, a lens, a deformable lens, a microelectromechanical systems mirror, an attenuator, a controlling optics, an optical filter, an amplitude modulator in a laser beam generator, or other suitable components.

(236) With reference now to FIG. 22, an illustration of a flowchart for controlling the emission of the laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 21 is yet another example of an implementation for operation **2002** in FIG. 20.

(237) The process emits the laser beams at the set of optical breakdown points causing the optical breakdowns that generate the radio frequency noise signals encoding the data (operation **2200**). The process terminates thereafter.

(238) Turning next to FIG. 23, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 23 is example of an implementation for operation **2200** in FIG. 22.

(239) The process begins by emitting the laser beams at a selected optical breakdown point in the set of optical breakdown points (operation **2300**). The process selects a new optical breakdown point in the set of optical breakdown points as the selected optical breakdown point in response to a set of optical breakdowns occurring at the selected optical breakdown point (operation **2302**).

(240) The process repeats emitting the set of laser beams and selecting a new optical breakdown point while generating the radio frequency noise signals encoding the data (operation **2304**) the process terminates thereafter. In operation **2304**, the process repeats operations **2300** and operation **2302** any number of times while transmitting the radio frequency noise signals. Operation at **2304** enables transmitting the radio frequency signals from different locations through the selection of different optical breakdown points. As result, identifying the origination of the radio frequency signals can be made more difficult.

(241) With reference next to FIG. 24, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 24 is another example of an implementation for operation **2002** in FIG. 20.

(242) The process begins by emitting a subset of the laser beams at the set of optical breakdown points to cause the optical breakdowns that generate the radio frequency noise signals encoding the data (operation **2400**). The process selects a new subset of laser beams as the subset of the laser beams (operation **2402**).

(243) The process repeats emitting the subset of laser beams and selecting a new subset of laser beams while transmitting the radio frequency noise signals encoding the data (operation **2404**). The process terminates thereafter. By using different subsets of the laser beams, identifying a location from which the laser beams originate can be made more difficult when the laser beams are emitted from different locations.

(244) In FIG. 25, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. 25 is an example of an implementation for operation **2002** in FIG. 20.

(245) The process emits the set of laser beams from different locations at an optical breakdown point, wherein a portion of the set of laser beams intersect at the optical breakdown point such that a power from the portion of the laser beams is sufficient to cause the optical breakdowns at the intersection point that generate the radio frequency noise signals encoding the data (operation **2500**). The process terminates thereafter.

(246) With reference to FIG. **26**, an illustration of a flowchart for controlling the emission of laser beams is depicted in points with an illustrative embodiment. The process illustrated in FIG. **26** is an example of an implementation for operation **2002** in FIG. **20**.

(247) The process emits the laser beams at an optical breakdown point (operation **2600**). The process terminates thereafter. In operation **2600**, the optical breakdowns occur in response to all of the laser beams intersecting at the optical breakdown point.

(248) With reference now to FIG. **27**, an illustration a flowchart for controlling laser beams is depicted in accordance with an illustrative embodiment. The process illustrated in this figure is an example of an additional operation that can be performed with the operations in FIG. **20**.

(249) The process changes a set of laser beam parameters for the laser beams to encode the data into the radio frequency noise signals (operation **2700**). The process terminates thereafter. In operation of **2700**, changing the set of laser beam parameters changes a set of radio frequency characteristics for the radio frequency noise signals. The set of radio frequency characteristics is selected from at least one of a timing, an optical breakdown point, an amplitude of the radio frequency noise signals, or other characteristics of the radio frequency noise signals.

(250) In this illustrative example, FIGS. **28-35** are flowcharts illustrating operations that can be performed to encode data in noise signals. Turning first to FIG. **28**, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. **28** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager **912** in computer system **910** in FIG. **9**.

(251) The process begins by identifying data for transmission (operation **2800**). The process transmits pulses of noise signals encoding the data (operation **2802**). The process terminates thereafter. The pulses of noise signals can be selected from at least one of electromagnetic frequency signals, radio frequency signals, microwave frequency signals, audio frequency signals, ultrasonic frequency signals, ultra-low frequency signals, very low frequency signals, underwater frequency signals, or optical frequency signals.

(252) In operation **2802**, the pulses of radio frequency noise signals can be transmitted in a number of different ways. For example, these pulses of noise signals can be radio frequency noise signals transmitted from a physical antenna. In another illustrative example, the pulses of noise signals can be transmitted using optical breakdowns generated by laser beams. The optical breakdowns can be controlled to generate pulses of noise signals in the form of radio frequency noise signals.

(253) The noise signals can be generated using at least one of a laser generation system that emits lasers to cause optical breakdown that generates the noise signal or an electric noise generator. The noise in the noise signal can be selected from at least one of nondeterministic noise or pseudo random noise.

(254) Turning to FIG. **29**, an illustration of a flowchart for transmitting pulses of noise signals is depicted in accordance with an illustrative embodiment. This flowchart is an example of an implementation for operation **2802** in FIG. **28**. In this example, the pulses of noise signals can be pulses of radio frequency noise signals.

(255) The process controls emission of a set of laser beams from a laser beam generator to cause optical breakdowns that generate the pulses of the radio frequency noise signals that encode the data (operation **2900**). The process terminates thereafter.

(256) With reference next to FIG. **30**, another illustration of a flowchart for transmitting pulses of

noise signals is depicted in accordance with an illustrative embodiment. This flowchart is an example of an implementation for operation **2802** in FIG. **28**.

(257) The process begins by generating a carrier radio frequency noise signal (operation **3000**). The process modulates the carrier noise signal to form the pulses of the noise signals (operation **3002**). In operation **3002**, the pulses encode the data.

(258) The process transmits the pulses of noise signals (operation **3004**). The process terminates thereafter.

(259) Turning now to FIG. **31**, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. **31** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager **912** in computer system **910** in FIG. **9**.

(260) The process begins by identifying data for transmission (operation **3100**). The process controls emission of a set of laser beams to cause optical breakdown that generate pulses of radio frequency noise signals (operation **3102**). The process terminates thereafter. In operation **3100**, the data can be encoded in the pulses of the radio frequency noise signals.

(261) With reference to FIG. **32**, an illustration of a flowchart of a process for controlling the mission of a set of laser beams is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for operation **3102** in FIG. **31**.

(262) The process controls a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the pulses of radio frequency noise signals encoding the data (operation **3200**). The process terminates thereafter.

(263) Turning next to FIG. **33**, an illustration of a flowchart of a process for controlling the emission of a set of laser beams is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for operation **3102** in FIG. **31**.

(264) The process controls the controlling emission of the set of laser beams to intersect an intersection point such that the power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data (operation **3300**). The process terminates thereafter.

(265) In FIG. **34**, an illustration of a flowchart for communicating data is depicted in accordance with an illustrative embodiment. The process in FIG. **34** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that is run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in communications manager **912** in computer system **910** in FIG. **9**.

(266) The process begins by receiving pulses of noise signals (operation **3400**). In operation **3400**, data is encoded in the pulses of noise signals.

(267) The process decodes the data encoded in the pulses of the noise signals using a set of characteristics of the pulses of the noise signals (operation **3402**). The process terminates thereafter. In operation **3402**, the set of characteristics comprises at least one of a timing of the pulses of noise, an amplitude of the pulses of noise, a duration of the pulses of noise, or some other characteristic.

(268) With reference now to FIG. **35**, an illustration of a flowchart for decoding data is depicted in accordance with an illustrative embodiment. The process depicted in this flowchart is an example of an implementation for operation **3402** in FIG. **34**.

(269) The process begins by receiving signals in a frequency range that includes the pulses of the noise signals encoding the data (operation **3500**). In operation **3500**, the signals in frequency range can be received using at least one of a bandpass filter, a notch filter, a band reject filters

(270) The process identifies the pulses of the noise signals in the frequency range (operation **3502**). The process terminates thereafter. In operation **3502**, the pulses of the noise signals in the

frequency range can be identified using an envelope detector.

(271) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program instructions, hardware, or a combination of the program instructions and hardware. When implemented in hardware, the hardware can, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program instructions and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program instructions run by the special purpose hardware.

(272) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

(273) Thus, the illustrative examples provide a method, apparatus, and system for transmitting radio frequency signals using a transmission system in which a physical antenna is absent. Optical breakdowns are generated by laser beams in which the optical breakdowns create plasma. The plasma results in radio frequency noise signals. The optical breakdowns can be controlled to encode data in the radio frequency noise signals. The locations of these optical breakdowns are radio frequency source emitters in the depicted examples.

(274) Further, these radio frequency source emitters can be moved to different locations by repositioning the laser beams such that the laser beams point at different optical breakdown points. Attacks at these locations are in essence attacks at the optical breakdown points where the plasma is generated.

(275) As a result, kinetic attacks against these locations are useless because no physical infrastructure is present at the locations. Further, the laser modulation sources are remote from the locations of these radio frequency source emitters. These optical breakdowns can occur at a location that is remote from the laser source.

(276) Further, the illustrative examples can encode data using noise signals. The use of noise signals is in contrast to the use of sinusoidal signals as a carrier signal to encode data. With the encoding of data in pulses of noise signals, issues with detection and interference in transmitting data encoded using sinusoidal carriers can be reduced.

(277) The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component can be configured to perform the action or operation described. For example, the component can have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Further, to the extent that terms “includes”, “including”, “has”, “contains”, and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term “comprises” as an open transition word without precluding any additional or other elements.

(278) Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable

others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A communications system comprising: a communications manager configured to: identify data for transmission; and transmit pulses of noise signals encoding the data, wherein the noise in the noise signals is nondeterministic noise; wherein the data is encoded in the pulse of the noise signals using at least one of a timing of the pulses and a duration of the pulses.
2. The communications system of claim 1, wherein the noise signals are selected from at least one of electromagnetic frequency noise signals, radio frequency noise signals, microwave frequency signals, audio frequency noise signals, ultrasonic frequency noise signals, ultra-low frequency noise signals, very low frequency noise signals, underwater frequency noise signals, or optical frequency noise signals.
3. The communications system of claim 1, wherein the pulses of the noise signals are pulses of radio frequency noise signals and further comprising: a laser generation system configured to emit a set of laser beams, wherein in transmitting the pulses of the noise signals the data, the communications manager is configured to: control emission of a set of laser beams from the laser generation system to cause optical breakdowns that generate the pulses of the radio frequency noise signals that encode the data.
4. The communications system of claim 1 further comprising: an electrical noise generator configured to generate a carrier frequency noise signal; and a modulator; and a transmitter, wherein in generating noise signals in pulses, the communications manager is configured to: generate a carrier noise signal using the electrical noise generator; and modulate the carrier noise signal using the modulator to form the pulses of the noise signals, wherein the pulses of the noise signals encode the data.
5. A communications system comprising: a laser generation system configured to emit a set of laser beams; a computer system; and a communications manager in the computer system, wherein the communications manager is configured to: identify data for transmission; and control an emission of the set of laser beams to cause optical breakdowns that generate pulses of radio frequency noise signals, wherein the data is encoded in the pulses of the radio frequency noise signals, wherein the noise in the noise signals is nondeterministic noise; wherein the data is encoded in the pulses of the radio frequency noise signals using at least one of a timing of the pulses and a duration of the pulses.
6. The communications system of claim 5, wherein in controlling the emission of the set of laser beams, the communications manager is configured to: control a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data.
7. The communications system of claim 5, wherein in controlling the emission of the set of laser beams, wherein the communications manager is configured to: control the emission of the set of laser beams to intersect an intersection point such that a power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data.
8. A method for communicating data, the method comprising: identifying data for transmission; and transmitting pulses of noise signals encoding the data, wherein the noise in the noise signals is nondeterministic noise; wherein the data is encoded in the pulses of the noise signals using at least one of a timing of the pulses of the noise signals and a duration of the pulses of the noise signals.
9. The method of claim 8 further comprising: encoding the data in the pulses of the noise signals by controlling at least one of a timing of the pulses and a duration of the pulses.
10. The method of claim 8, wherein the noise signals are selected from at least one of

electromagnetic frequency noise signals, radio frequency noise signals, microwave frequency signals, audio frequency noise signals, ultrasonic frequency noise signals, ultra-low frequency noise signals, very low frequency noise signals, underwater frequency noise signals, or optical frequency noise signals.

11. The method of claim 8, wherein the pulses of the noise signals are pulses of radio frequency noise signals and wherein transmitting the pulses of the noise signals comprises: controlling emission of a set of laser beams from a laser beam generator to cause optical breakdowns that generate the pulses of the radio frequency noise signals that encode the data.

12. The method of claim 8, wherein transmitting the pulses of the noise signals encoding the data comprises: generating a carrier radio frequency noise signal; and modulating the carrier radio frequency noise signal to form the pulses of the noise signals, wherein the pulses encode the data; and transmitting the pulses of the noise signals.

13. A method for communicating data, the method comprising: identifying data for transmission; and controlling emission of a set of laser beams to cause optical breakdowns that generate pulses of radio frequency noise signals, wherein the data is encoded in the pulses of the radio frequency noise signals, wherein the noise in the noise signals is nondeterministic noise; wherein the data is encoded in the pulses of the radio frequency noise signals using at least one of a timing of the pulses of the radio frequency noise signals and a duration of the pulses of the radio frequency noise signals.

14. The method of claim 13, wherein controlling the emission of the set of laser beams comprises: controlling a power of a laser beam in the set of laser beams to reach an optical breakdown level at a focal point to cause the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data.

15. The method of claim 13, wherein controlling the emission of the set of laser beams comprises: controlling the emission of the set of laser beams to intersect an intersection point such that a power of the set of the laser beams at the intersection point causes the optical breakdowns that generate the pulses of the radio frequency noise signals encoding the data.

16. A method for communicating data, the method comprising: receiving pulses of noise signals, wherein data is encoded in the pulses of the noise signals; and decoding the data encoded in the pulses of the noise signals using a set of characteristics of the pulses of the noise signals, wherein the noise in the noise signals is nondeterministic noise; wherein the data is encoded in the pulses of the noise signals using at least one of a timing of the pulses of the noise signals and a duration of the pulses of the noise signals.

17. The method of claim 16, wherein the set of characteristics comprises at least one of a timing of the pulses of noise and a duration of the pulses of noise.

18. The method of claim 17, wherein decoding the data encoded in the pulses of the noise signals comprises: receiving signals in a frequency range that includes the pulses of the noise signals encoding the data; and identifying pulses of the noise signals in the frequency range.

19. The method of claim 18, wherein the frequency range is received using at least one of a bandpass filter, a notch filter, or a band reject filter.

20. The method of claim 18, wherein the pulses of the noise signals in the frequency range are identified using an envelope detector or a clipper circuit.
